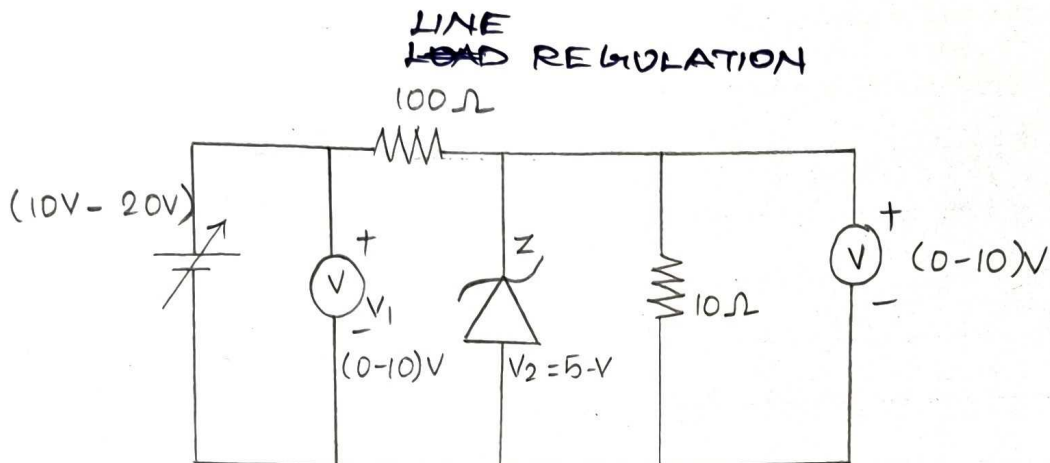


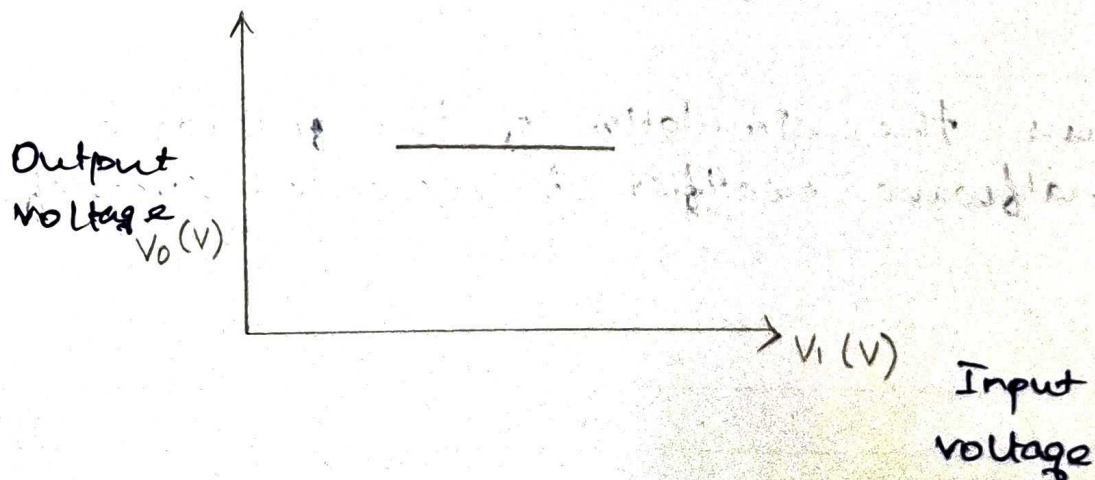
CIRCUIT DIAGRAM:



TABULATION

S.No	INPUT VOLTAGE V_1 (V)	OUTPUT VOLTAGE V_0 (V)
1	+12.0	+5.14
2	+13.0	+5.16
3	+14.0	+5.17
4	+15.0	+5.17
5	+16.0	+5.18
6	+17.0	+5.19

MODEL GRAPH



ZENER DIODE AS VOLTAGE REGULATION

AIM :

To simulate the line and load regulation operation zener diode.

APPARATUS REQUIRED :

Laptop with proteus software.

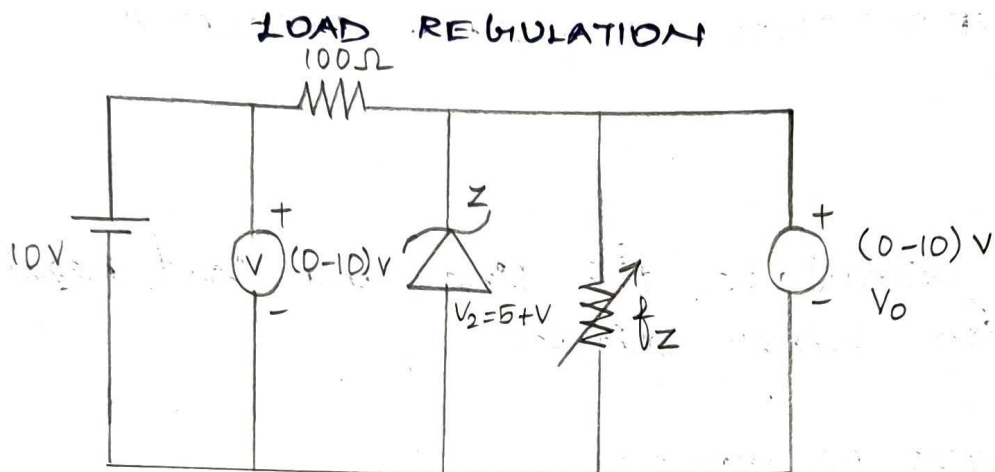
THEORY :

A heavily doped PN junction diode is called zener diode. Due to heavily doped nature zener diode works under forward bias and reverse bias conditions.

Zener diode under reverse bias condition is used as a voltage regulation. Even if the input voltage changes the output is constant. This is called line regulation.

If the load resistance changes the output voltage is constant this is called load regulation.

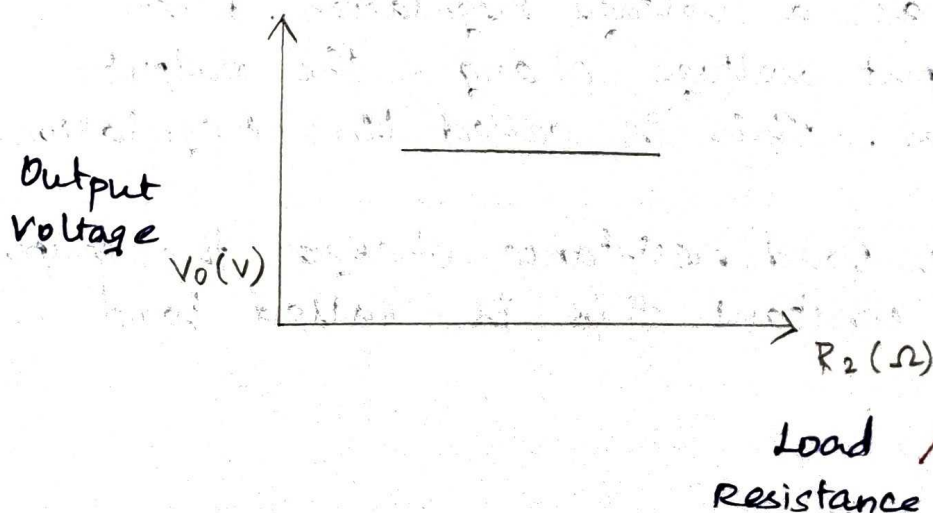
CIRCUIT DIAGRAM



TABULATION :

S.No	LOAD RESISTANCE $R_2 (\Omega)$	OUTPUT VOLTAGE $V_o (V)$
1	100 Ω	+5.19
2	200 Ω	+5.20
3	300 Ω	+5.20
4	400 Ω	+5.20
5	500 Ω	+5.20
6	600 Ω	+5.21

MODEL GRAPH



PROCEDURE :

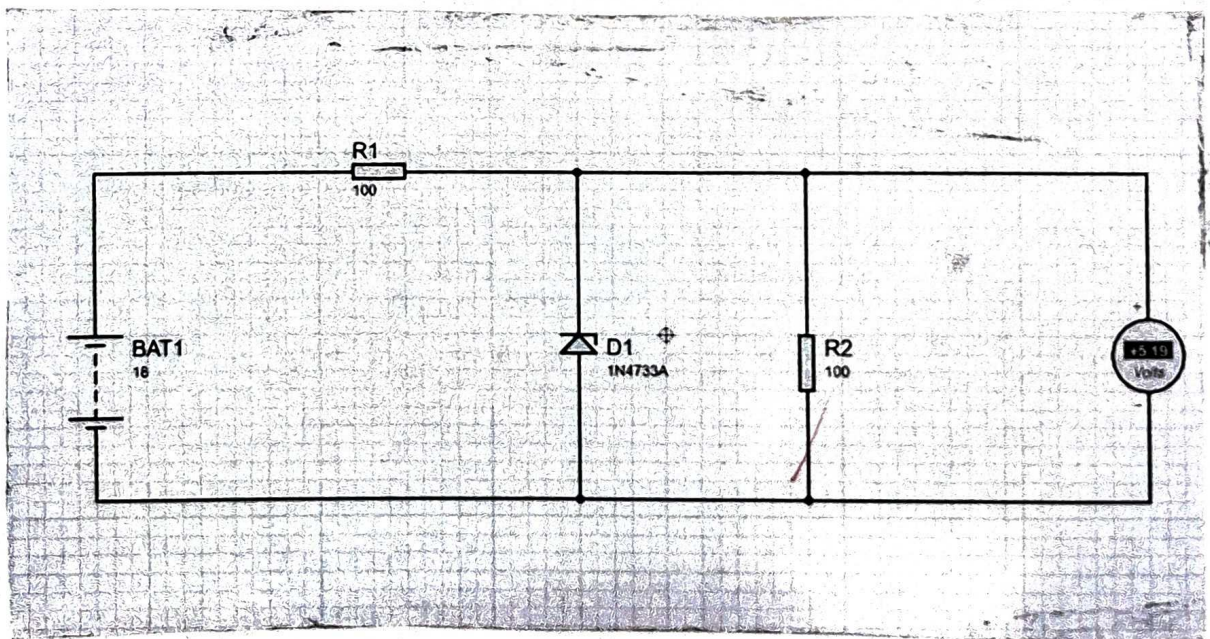
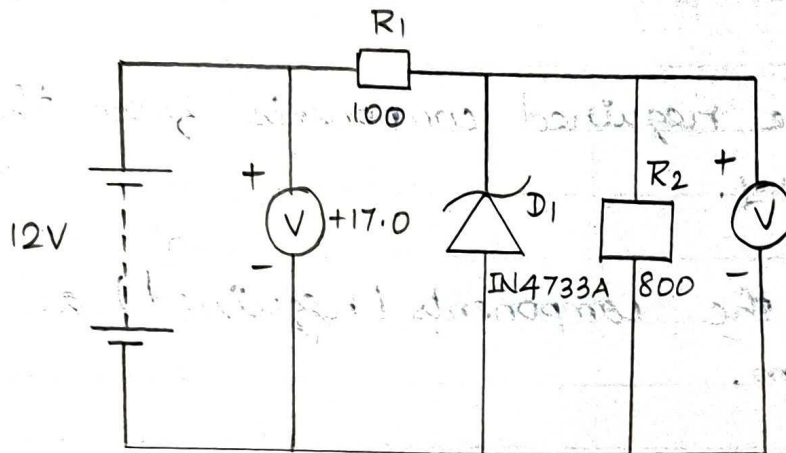
* Drag the required components from the proteus library.

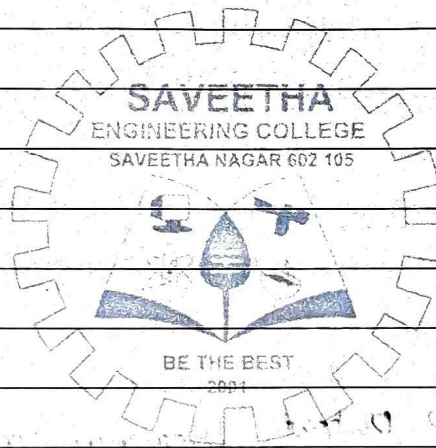
* Connect the components (required) as per circuit diagram.

* For line regulation vary the input voltage and not the output voltages in tabulation. Draw the graph between input voltage and output voltage.

* For load regulations vary the load resistance and note the output voltage in tabulations. Draw the graph between load resistance and output voltage.

OUTPUT:





RESULT:

Thus simulations of line load regulation operation of Zener diode successfully completed.